

SESSION 3

CQs:

1. Which specific instances of **RoadSign** have the **hasShape** property linking them to the **shape_circle** instance?
2. What is the integer value (**signValue**) associated with the mandatory speed limit sign **Sign_D4**?
3. Which images in the dataset have an **isPano** data property set to the boolean value of **true**?
4. For a given traffic sign object, what are the exact decimal values of its **xmin** and **ymax** data properties?
5. If a user attempts to assign two different **width** integers to the same image, will the ontology reasoner flag it as an inconsistency due to the property's **Functional** characteristic?

LLM: Gemini

LLM Prompt: Role: You are an expert Ontology Engineer specialising in the domain of traffic sign knowledge graphs. **Objective:** Generate OWL2 RDF/XML code to create owl:DatatypeProperty elements for a traffic sign ontology. **Scope & Domain:**

Domain: The Vienna Convention on Road Signs and the Mapillary Traffic Sign Dataset (MTSD).

Key Entities: We need data properties for sign values (e.g., speed limits, weight limits as integers) and MTSD metadata (e.g., mtsd:width as integer, mtsd:isPano as boolean, mtsd:xmin as decimal, mtsd:occluded as boolean).

Boundaries: DO NOT generate Object Properties or Classes. ONLY generate Data Properties with their appropriate rdfs:domain and rdfs:range (xsd:integer, xsd:decimal, xsd:boolean, xsd:string). **Competency Questions:** The ontology must be able to answer the following questions:

What are the exact bounding box coordinates (xmin, ymin, xmax, ymax) for a given traffic sign object?

What are the pixel dimensions (width and height) of a source image?

What is the specific integer value (e.g., 50, 70) of a speed limit sign?

Structural Requirements: Define each `<owl:DatatypeProperty>`. Assign the appropriate `rdfs:domain` (e.g., `RoadSign` or `Image`) and `rdfs:range` (e.g., `xsd:integer`, `xsd:boolean`). Use <http://www.ontology.org/traffic/vienna-convention#> as the base namespace. **Output Format:** Provide ONLY the raw RDF/XML code snippet containing the new data property declarations. Do not write explanations.